

Solution Requirements

Date	29 September 2026
Team ID	SWTID-2026-7019
Project Name	EduGenie - AI Learning Assistant
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	The current release is an open learning tool with no student login or stored accounts. Access to the paid AI service is protected server-side: the Gemini API key is kept in the .env file, is never sent to the browser, and every AI endpoint checks that a valid key is configured before processing (otherwise a JSON error is returned). Planned enhancement: sign-in with Google OAuth so students can save their history.	Medium
2	Authorization levels	Learner / Student: can use all five features (ask a question, get an explanation, summarize text, take a quiz, get a learning path). Administrator / Developer: manages the .env configuration (API key, Gemini model, local model), deployment and CORS settings. Teacher role (view class progress) is a future extension.	Medium
3	External interfaces	1) Google Gemini API via the google-genai SDK for Q&A, summaries, quizzes and learning paths. 2) Hugging Face Hub, to download the MBZUAI/LaMini-Flan-T5-783M model used locally for explanations. 3) Internal REST endpoints used by the browser: GET /qa, POST /explain/, POST /summarize/, POST /quiz and GET /learn/recommendations.	High

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
4	Transactions processing	Each feature is a stateless request-response transaction: the browser sends input with the Fetch API, FastAPI validates it, builds a prompt, calls Gemini (or the local model), and returns JSON. Quiz output is cleaned of Markdown code fences and parsed into a list of questions, options and answers. Failures return a JSON error (HTTP 400 or 500) that is shown to the student. No payments are involved.	High
5	Reporting	No stored reports. Results are shown on screen: answers, explanations, summaries and learning paths (beginner, intermediate and advanced levels with resources). The quiz report shows correct/wrong answers per question and a final score such as 2/3 with an encouragement message.	Medium
6	Business rules, etc.	Empty input is rejected with a message. A quiz has exactly 3 multiple-choice questions, each with 4 options and one correct answer that matches an option. Each question can be answered only once. Explanations must be simple enough for a school student and summaries in simple language. Score message: perfect score, at least 50% (good job), otherwise keep practicing.	High
7	Compliance to laws or regulations	Store no personal data and no student accounts (data minimisation). Text entered by users is sent to the Google Gemini API and is subject to Google's API terms; users should not enter sensitive personal data. Follow the principles of India's DPDP Act 2023 and GDPR, keep API keys out of version control, and respect the open-source licences of FastAPI, Transformers and PyTorch.	Medium
8	Other	Loading state on buttons while a request runs, toast messages for input errors, Enter-key submit for text fields, and automatic fallback to Gemini when the local explanation model is unavailable.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The interface loads quickly and gives immediate feedback; AI responses depend on Gemini latency, so a loading indicator is shown until the result arrives. Local explanation output is limited to 150 new tokens to keep it fast.	Page load < 2 seconds; input errors shown instantly; AI response within about 10 seconds for 90% of requests
2	Scalability	The backend is stateless (no session or database), so more Uvicorn instances can be run behind a load balancer as the number of students grows. Gemini API quota is the main limit.	Supports at least 100 concurrent users on a single instance; scales horizontally

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3	Security & Data Privacy	API key stored only in .env and excluded from Git; never exposed to the frontend. CORS restricted to allowed origins; input validated on the server; HTTPS in production; no personal data stored.	0 secrets in the repository or browser responses; 100% of production traffic over HTTPS (TLS 1.2+)
4	Reliability & Availability	A global exception handler always returns JSON errors instead of crashing; explanations fall back to Gemini if the local model fails; the app can be restarted with one command.	99% uptime per month; every failed request returns a readable error message
5	Usability & Accessibility	Simple single-page layout with one clear card per feature, responsive design, readable colours, keyboard support (Enter key) and clear success/error messages.	New user completes any feature in under 1 minute; aims to meet WCAG 2.1 AA; works on latest Chrome, Edge, Firefox
6	Other	Maintainability: one Python module per feature (qna, explanation, summary, quiz, learning path) and dependencies listed in requirements.txt. Portability: model names and keys are configurable through environment variables and run on Windows, Linux and macOS.	Adding a new feature requires only one new module and one route; setup in under 10 minutes